

Kentstar Samuel Harsono

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EDUCATION

University of California, Irvine

Bachelor of Science in Biomedical Engineering, Minor in CS, Statistics, Biophotonics Specialization

- Relevant Coursework: Finite Element Analysis, Computer-Aided Design, Statistical Analysis

Tohoku University

Department of Electrical, Information and Physics Engineering, Junior Year Program in English

- Relevant Coursework: Stress Analysis, Data Structures and Algorithms, Disaster Prevention

American EMT Academy

Emergency Medical Technician Certification

- Relevant Coursework: NREMT Patient Assessment, Medical Device Training

Norco College

Associate of Science: Math and Science Great Distinction, Certification in Physics

- Relevant Coursework: Mathematical Analysis, Thermodynamics, Circuit Design

PRODUCTS & PROJECTS

LCD-Inspired Spatial Light Modulator | Soldering, Electronics, SolidWorks, Arduino

Spring 2025

- Developed LCD-inspired SLM using custom electronics and compartmental design in SolidWorks, enabling dynamic phase shifting for laser applications while minimizing system cost and simplifying experimental integration

Vortex Spiral Phase Plate | SolidWorks, MATLAB

Spring 2025

- Designed and developed a reusable spiral plate mold for the precise fabrication of phase plates, optimizing the process for optical trapping of nanoparticles and removing additional cost offsets to laboratory

Pedestrian Dynamical Friendship Model | Python, C++, Linux

Spring 2024

- Collaborated with professor to simulate and refine a universal pedestrian dynamics models, inspired by Japan's Shibuya Scramble Crossing

Microscopic Stage | Python, SolidWorks, Arduino, ImageJ

Spring 2023

- Invented and analyzed efficiency of cost-efficient micromechanical stepper system component with open-source code for lithography applications, achieving micrometer precision and reducing system costs by at least 500%

Adjustable Hearing Aid | SolidWorks

Spring 2023

- Led a team of 5 members to design a 3D-printed hearing aid for diverse ear types, using SolidWorks and finite element analysis to enhance structural integrity and electrical spacing, ensuring optimal patient comfort and usability

Heart Rate Monitor | MATLAB, SolidWorks, LabVIEW, Arduino

Winter 2022

- Collaborated with 8 members to design LabVIEW-based reflective optomechanical system for wrist applications used in modern watch technology via signal processing and integration of I2C communication with BH1750 photosensor

EXPERIENCE

New World Medical

Research and Development, Intern

Rancho Cucamonga, CA

Sep 2025 - Present

- Developed optical fiber alignment systems to enhance coupling efficiency in glaucoma surgical devices, improving device calibration accuracy by optimizing mechanical and optical alignment
- Designed and prototyped SolidWorks components integrating optical and thermal elements to boost device reliability

Beckman Laser Institute & Medical Clinic

Undergraduate Researcher

Irvine, CA

Jan 2023 - Present

- Conducted and optimized low-cost lithographic techniques for 10+ hours weekly under PI Dr. Daryl Preece at UCI's Quantum Biophotonics Lab, achieving sub-micron precision on par with industrial standards
- Integrating low-budgeted systems to eliminate subscription-focused softwares and over budgeted systems in the lab

Tohoku University / Natsu-Lino

Project Marketing

Sendai, Japan

April 2024 - Aug 2024

- Conducted research on gelato production to develop country-inspired flavors for Natsu-Lino

- Increased market sales by 5% by networking with social media food celebrities and television network across Sendai

- Facilitated the venue setup to ensure a smooth and efficient event execution

Research Institute of Electrical Communication Tohoku University

Special-Appointed Undergraduate Researcher

Sendai, Japan

Sep 2023 - Aug 2024

- Conducted research in Japan on decentralized pedestrian dynamics to enhance modern movement modeling under PIs Dr. Takeshi Kano and Dr. Akio Ishiguro

- Presenting findings through poster conferences and seminars to academic / international audiences and faculty